

Project

Secure Coding Review

1.Objective

The objective of this project is to identify security vulnerabilities in a simple Python login program and recommend secure coding practices.

1.Vulnerability - 01

Hardcoded Password

Risk

The password is directly stored in the source code.

Anyone with access to the code can see it.

Solution

Store passwords securely using environment variables or databases.

2.Vulnerability – 02

Plain Text Password

Risk

Passwords are stored without encryption.

If the code is leaked, passwords are exposed.

Solution

Use password hashing algorithms like bcrypt or Argon2.

3.Vulnerability – 03

No Input Validation

Risk

The application accepts any user input.

This can lead to unexpected behavior.

Solution

Validate and sanitize all user input.

4.Vulnerability – 04

Unlimited Login Attempts

Risk

Attackers can perform brute-force attacks.

Solution

Limit failed login attempts and implement account logout.

5.Vulnerability – 05

Lack of Logging

Risk

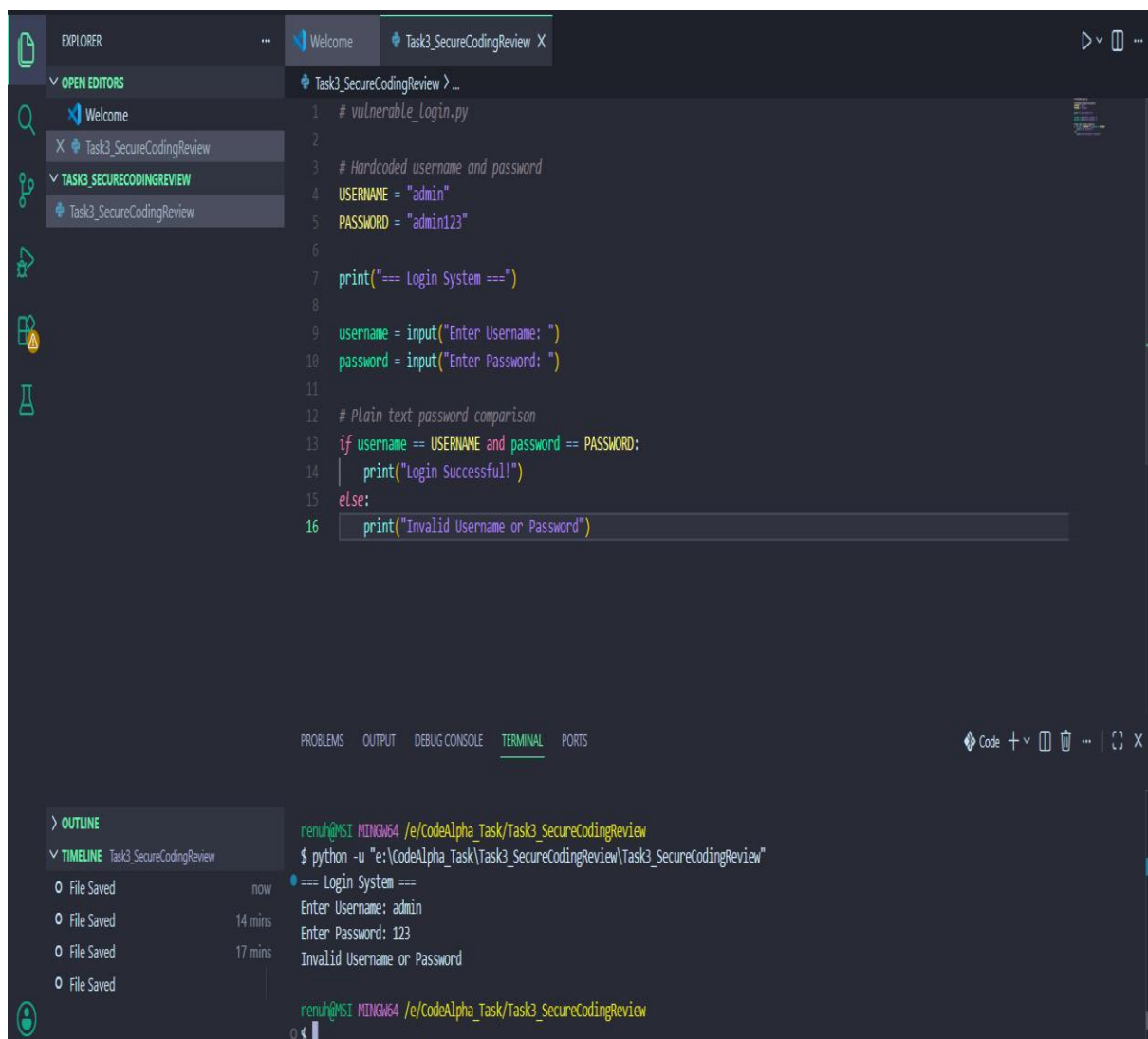
Security incidents cannot be monitored.

Solution

Maintain secure authentication logs.

Conclusion

The application contains several intentional security vulnerabilities. Following secure coding practices such as password hashing, input validation, secure credential storage, and login attempt restrictions significantly improves application security.



The screenshot displays a Visual Studio Code editor with a Python file named `Task3_SecureCodingReview.py`. The script implements a login system with several intentional security vulnerabilities: it uses hardcoded credentials, performs a plain text password comparison, and lacks input validation and login attempt restrictions. Below the editor, a terminal window shows the execution of the script, demonstrating that the hardcoded credentials are accepted for login.

```
1 # vulnerable_login.py
2
3 # Hardcoded username and password
4 USERNAME = "admin"
5 PASSWORD = "admin123"
6
7 print("=== Login System ===")
8
9 username = input("Enter Username: ")
10 password = input("Enter Password: ")
11
12 # Plain text password comparison
13 if username == USERNAME and password == PASSWORD:
14     print("Login Successful!")
15 else:
16     print("Invalid Username or Password")
```

Terminal Output:

```
remuh@MSI MINGW64 /e/CodeAlpha_Task/Task3_SecureCodingReview
$ python -u "e:\CodeAlpha_Task\Task3_SecureCodingReview\Task3_SecureCodingReview"
=== Login System ===
Enter Username: admin
Enter Password: 123
Invalid Username or Password

remuh@MSI MINGW64 /e/CodeAlpha_Task/Task3_SecureCodingReview
$
```

The image shows a Visual Studio Code (VS Code) interface with a Python script open in the editor. The script is a simple login system with a hardcoded username and password. Below the editor, the terminal shows the command to run the script and its output.

EXPLORER

- OPEN EDITORS
 - Task3_SecureCodingReview
- TASK3_SECURECODINGREVIEW
 - Task3_SecureCodingReview

EDITOR

```
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3 # Hardcoded username and password
4 USERNAME = "admin"
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7 print("=== Login System ===")
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9 username = input("Enter Username: ")
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12 # Plain text password comparison
13 if username == USERNAME and password == PASSWORD:
14     print("Login Successful!")
15 else:
16     print("Invalid Username or Password")
```

TERMINAL

```
renuh@MSI MINGW64 /e/CodeAlpha_Task/Task3_SecureCodingReview
$ python -u "e:\CodeAlpha_Task\Task3_SecureCodingReview\Task3_SecureCodingReview"
=== Login System ===
Enter Username: admin
Enter Password: admin123
Login Successful!

renuh@MSI MINGW64 /e/CodeAlpha_Task/Task3_SecureCodingReview
$
```

OUTLINE

- TIMELINE Task3_SecureCodingReview
 - File Saved now
 - File Saved 14 mins
 - File Saved 17 mins
 - File Saved